Elasticity

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Static Equilibrium

- Equilibrium implies that the object moves with both constant velocity and constant angular velocity relative to an observer in an inertial reference frame.
- Will deal now with the special case in which both of these velocities are equal to zero
  - This is called *Static Equilibrium.*
- Static equilibrium is a common situation in engineering.
- The principles involved are of particular interest to civil engineers, architects, and mechanical engineers.

Elasticity

- An **elastic object** returns to its original shape when the deforming forces are removed.
- Various elastic constants will be defined, each corresponding to a different type of deformation.
- Actually, all objects are deformable to some extent.
  - It is possible to change the size and/or shape of the object by applying external forces.
- Internal forces resist the deformation.

Definitions Assoc. with Deformation

- **Stress**
  - Is proportional to the force causing the deformation
  - It is the external force acting on the object per unit cross-sectional area.

- **Strain**
  - Is the result of a stress
  - Is a measure of the degree of deformation

Elastic Modulus

- The elastic modulus is the constant of proportionality between the stress and the strain.
  - For sufficiently small stresses, the stress is directly proportional to the stress.
  - It depends on the material being deformed.
  - It also depends on the nature of the deformation.
- The elastic modulus, in general, relates what is done to a solid object to how that object responds.

$$\text{Elastic Modulus} \equiv \frac{\text{Stress}}{\text{Strain}}$$

- Various types of deformation have unique elastic moduli.

Types of Moduli

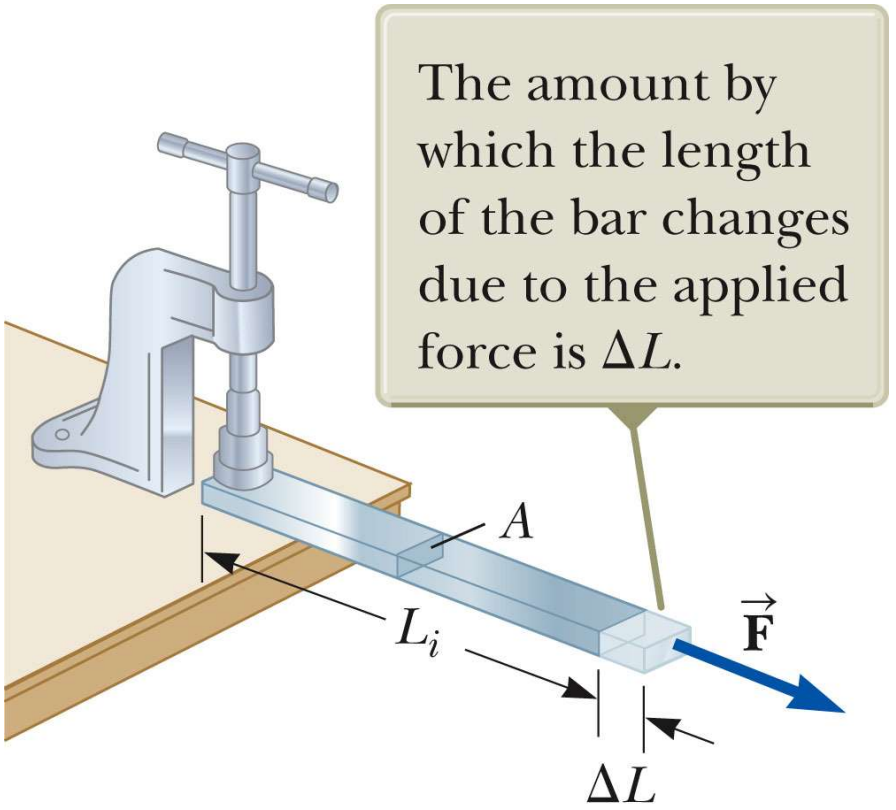
- **Young's Modulus**
  - Measures the resistance of a solid to a change in its length
- **Shear Modulus**
  - Measures the resistance of motion of the planes within a solid parallel to each other
- **Bulk Modulus**
  - Measures the resistance of solids or liquids to changes in their volume

Young's Modulus

- The bar is stretched by an amount ΔL under the action of the force F.

- The **tensile stress** is the ratio of the magnitude of the external force to the cross-sectional area A.

- The **tension strain** is the ratio of the change in length to the original length.

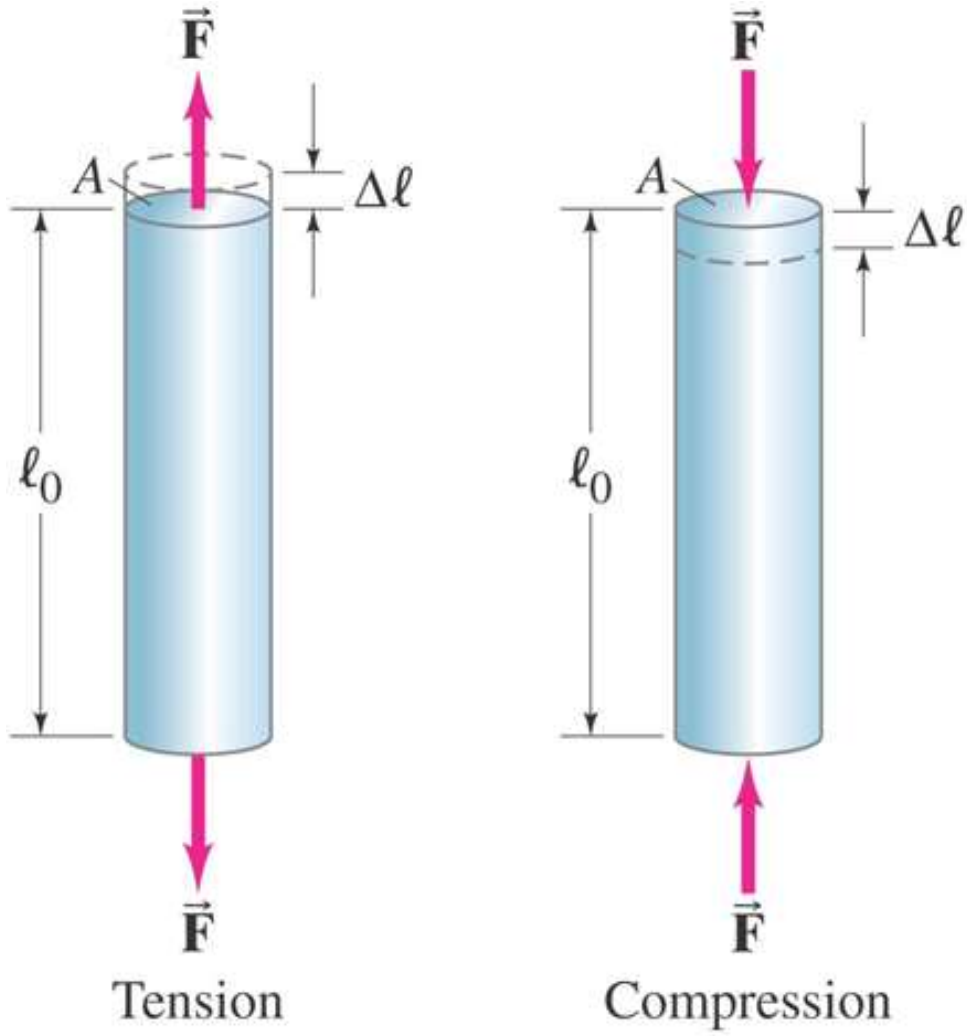


$$Y \equiv \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{F/A}{\Delta L/L_i}$$

Stress and Strain

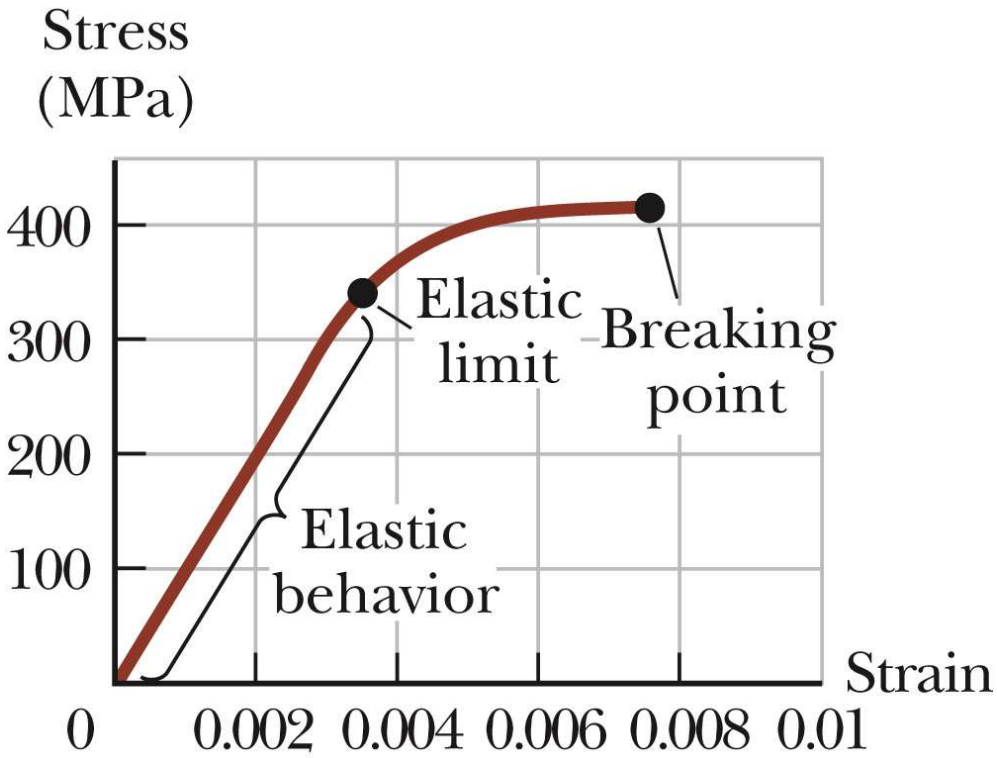
- In tensile stress, forces tend to stretch the object.
- In Pressure stress, forces tend to compress the object.

$$Y \equiv \frac{tensile\ stress}{tensile\ strain} = \frac{F/A}{\Delta L/L_o}$$



Stress vs. Strain Curve

- Experiments show that for certain stresses, the stress is directly proportional to the strain.
- This is the elastic behavior part of the curve.



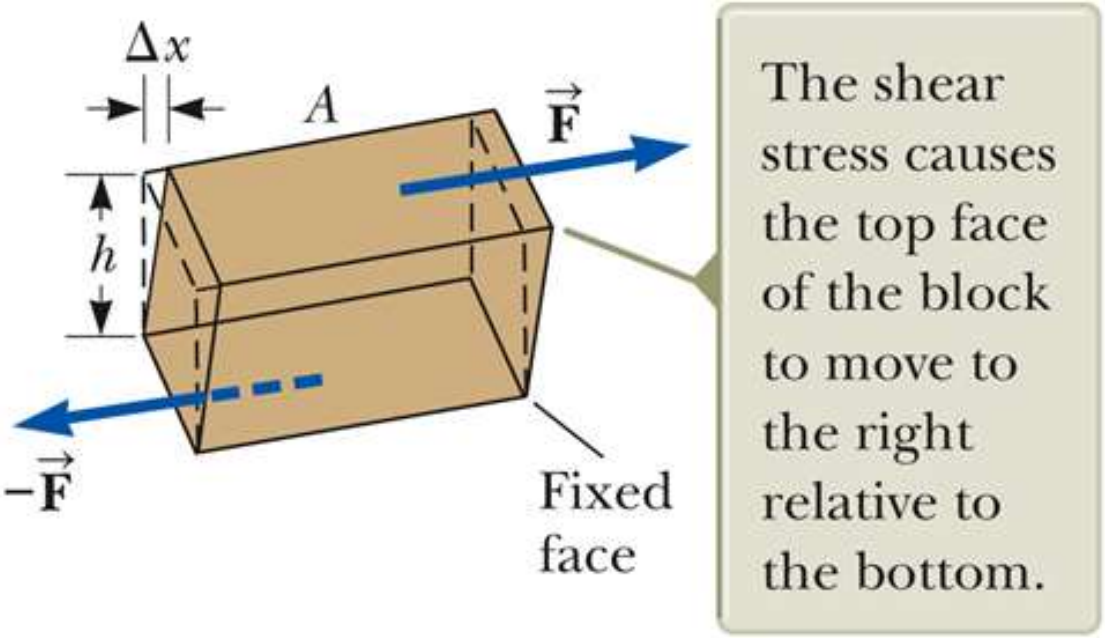
Elastic limit

is the maximum stress that can be applied to the substance before it becomes permanently deformed.

Shear Stress



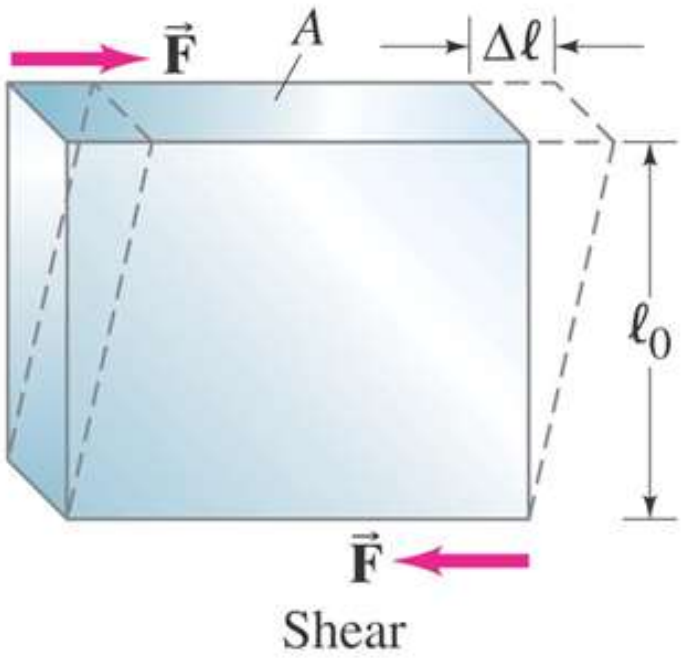




- The force acts <u>parallel</u> to one of its faces while the opposite face is held fixed by another force. It is called a ***shear stress.***

Stress on Rigid Body

- The shear strain is $\Delta l / L_o$.
  - Δl is the horizontal distance the sheared face moves.
  - L_o is the height of the object.
- The shear stress is F/A.
  - F is the tangential force.
  - A is the area of the face being sheared.
- The shear modulus is the ratio of the shear stress to the shear strain.

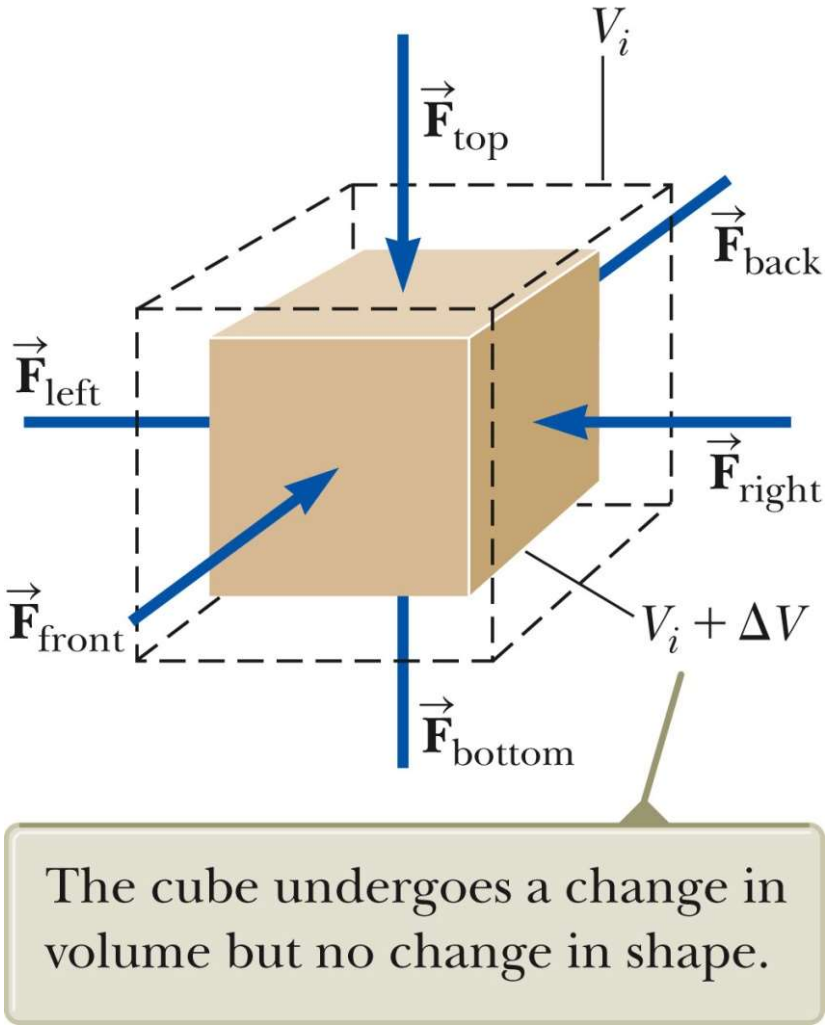


Shear

$$S = \frac{shear\ stress}{shear\ strain} = \frac{F/A}{\Delta l / Lo}$$

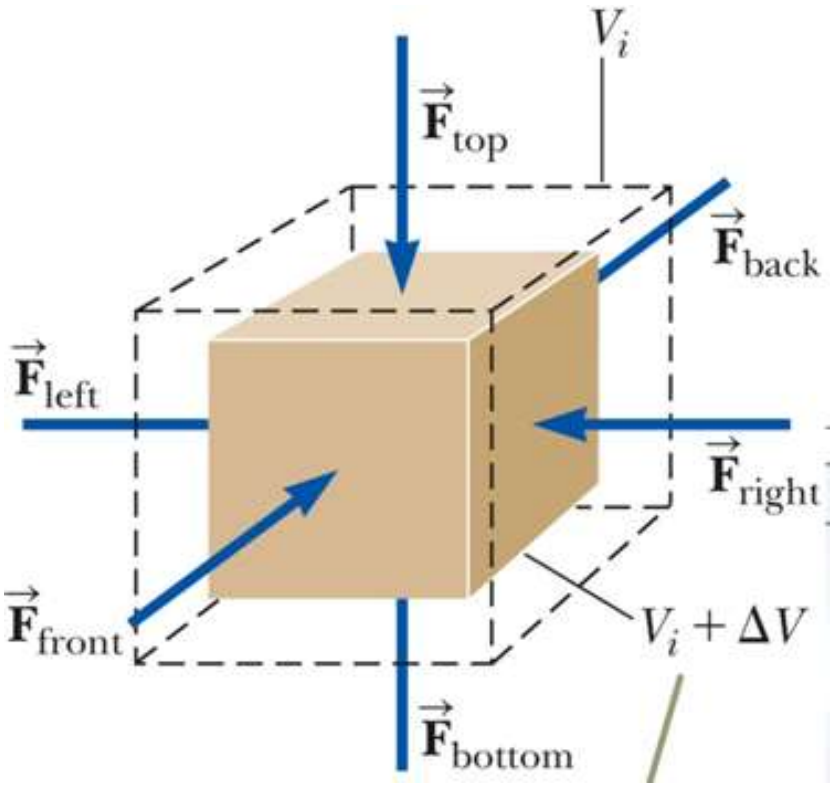
Bulk Modulus

- force of uniform magnitude is applied perpendicularly over the entire surface of the object.
- The object will undergo a change in volume, but not in shape.



Bulk Modulus

- The volume stress is defined as the ratio of the magnitude of the total force, *F*, exerted on the surface to the area, *A*, of the surface.
- The volume strain is the ratio of the change in volume to the original volume.
- The bulk modulus is the ratio of the volume stress to the volume strain.



$$B = \frac{\text{volume stress}}{\text{volume strain}} = -\frac{\Delta F/A}{\Delta V/V_i} = -\frac{\Delta P}{\Delta V/V_i}$$

Bulk Modulus

- Both solids and liquids have a bulk modulus.
- Liquids cannot sustain a shearing stress or a tensile stress.
  - If a shearing force or a tensile force is applied to a liquid, the liquid will flow in response.

Elastic Moduli

TABLE 12–1 Elastic Moduli

Material	Young's Modulus, E (N/m^2)	Shear Modulus, G (N/m^2)	Bulk Modulus, B (N/m^2)
Solids			
Iron, cast	100×10^9	40×10^9	90×10^9
Steel	200×10^9	80×10^9	140×10^9
Brass	100×10^9	35×10^9	80×10^9
Aluminum	70×10^9	25×10^9	70×10^9
Concrete	20×10^9		
Brick	14×10^9		
Marble	50×10^9		70×10^9
Granite	45×10^9		45×10^9
Wood (pine) (parallel to grain)	10×10^9		
(perpendicular to grain)	1×10^9		
Nylon	5×10^9		
Bone (limb)	15×10^9	80×10^9	
Liquids			
Water			2.0×10^9
Alcohol (ethyl)			1.0×10^9
Mercury			2.5×10^9
Gases[†]			
Air, H_2, He, CO_2			1.01×10^5

[†]At normal atmospheric pressure; no variation in temperature during process.

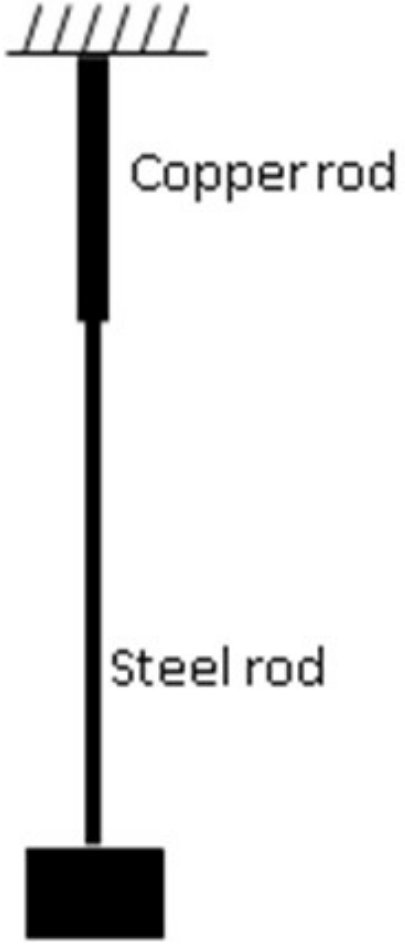
Example: 1

- Tension in piano wire
- A 1.60-m-long steel piano wire has a diameter of 0.20 cm. How great is the tension in the wire if it stretches 0.25 cm when tightened?

F = 980 N

Example: 2

- Two Rods of copper and steel are joined together and support the weight of a mass *m* of 50 kg as shown in the figure. The copper rod has a cross-sectional area of 0.080 cm^2 and a length of 3.0 m while the steel rod has a cross-sectional area of 0.040 cm^2 and a length of 6.0 m. Find the stress and elongation in each rod



$$Young's\ modulus\ for\ copper = 11x10^{10}\ N/m^2$$

$$Young's\ modulus\ for\ steel = 20x10^{10}\ N/m^2$$

Example: 3

A solid brass sphere is initially surrounded by air, and the air pressure exerted on it is 1.0×10^5 N/m^2 (normal atmospheric pressure). The sphere is lowered into the ocean to a depth where the pressure is 2.0×10^7 N/m^2. The volume of the sphere in air is 0.50 m^3. By how much does this volume change once the sphere is submerged?

$-1.6 * 10^{-4}\ m^3$

Example: 4

- How large a pressure increase (in ATM) must be applied to water if it is to be compressed in volume by 1%? The bulk modulus of water is 2×10^9 N/m^2 and 1 ATM = 10^5 N/m^2.

200 ATM

Example: 5

- An elevator of mass of 1000kg is hanged by steel wire with a cross sectional area $1.2*10^{-4}m^2$ (Young's modulus $2*10^{11}$ N/m^2). The strain in the wire should not exceed 0.001. If the mass of the maximum load that can be of the elevator is 1000kg,
- what is the maximum acceleration of the elevator while it is going up. (in m/s^2)
- What is the tension stress in the wire in (N/m^2)?

$2.2\ m/s^2$ & $2*10^8\ N/m^2$

Example: 6

- A walkway suspended across a hotel lobby is supported at numerous points along its edges by a vertical cable above each point and a vertical column underneath. The steel cable is 1.27 cm in diameter and is 5.75 m long before loading. The aluminum column is a hollow cylinder with an inside diameter of 16.14 cm, an outside diameter of 16.24 cm, and an unloaded length of 3.25 m. When the walkway exerts a load force of 8500 N on one of the support points, how much does the point move down?

8.60×10^{-4} m